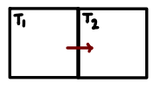
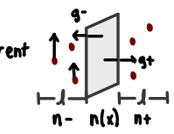
Diffusion



Classical Gas Case

- 'rms' velocity
  $\bar{v} = \sqrt{\sum_i |\vec{v}_i|}$
  $|\vec{v}| = v_{x,i}^2 + v_{y,i}^2 + v_{z,i}^2$
  $(\frac{1}{2}mv^2 = \frac{3}{2}k_BT)$
- average velocity
  $\vec{v} = \sum_i \vec{v}_i$
- mean free path
  $l = \vec{v}\tau$ ← scattering / relaxation / decoherence time*

- number current
  $\vec{g} = n\vec{v}$



$g_+ = \frac{1}{2}n_-v_x$
$g_- = \frac{1}{2}n_+v_x$
net flow:
$g = g_+ - g_-$
↳ $= \left(\frac{n_- - n_+}{2l}\right) l v_x$
$= -\frac{dn}{dl}$

$dx = 2l$
$n_+ = n(x+dx)$
$n_- = n(x-dx)$

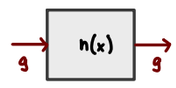
$\Rightarrow \vec{g} = -\frac{dn}{dx}\overbrace{l v_x}^{D}$

define D
→ diffusion const

→ $D = l v_x$
$= \bar{v}_x^2\tau$
$= \frac{1}{3}\bar{v}^2\tau = k_BT\frac{\tau}{m}$

where $\bar{v}^2 = \bar{v}_x^2 + \bar{v}_y^2 + \bar{v}_z^2$
$= 3\bar{v}_x^2$
→ $\bar{v}_x^2 = \bar{v}^2/3$

* $\frac{1}{2}mv^2 = \frac{3}{2}k_BT$
↳ $\bar{v}^2 = 3k_BT/m$



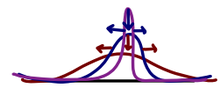
Continuity condition

$\frac{dn}{dt} = -\frac{dg}{dx}$, $\boxed{\frac{dn}{dx} = D\frac{d^2n}{dx^2}}$
diffusion equation

3D: $\frac{dn}{dt} = D\nabla^2 n$

* $g = -D\frac{dn}{dx}$
$= -D\vec{\nabla}n$

1D Solution: $n(x,t) = \frac{1}{(4\pi t)^{\frac{1}{2}}}\underbrace{e^{-x^2/4Dt}}_{\text{Gaussian}}$



spreads out

$\sigma^2 = \sqrt{4Dt}$
width

energy density: $u = \bar{E}n$
→ $\frac{du}{dt} = D\frac{d^2u}{dx}$ → $\frac{dT}{dt} = D\frac{d^2T}{dx}$

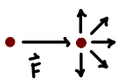
heat capacity:
$C = \frac{dU}{dT} = V\frac{du}{dT}$ → $du = \frac{C}{V}dT$

$\vec{h} = u\vec{v} = -D\vec{\nabla}n$
↑ avg. velocity
→ $= -D\frac{C}{V}\vec{\nabla}T$
$= -K\nabla T \Rightarrow K = \frac{D(C/V)}{} \rightarrow D = K\frac{V}{C}$
↑ heat conductivity

- $\vec{g} = n\vec{v}$
  $n\vec{v} = -D\frac{dn}{dx}$
  $\vec{v} = -D\frac{1}{n}\frac{dn}{dx}$

recall, $n(x) = n_0e^{\mu}$

Drift under $\vec{F}$ w/ random scattering



$\vec{F} = ma = m\frac{d\vec{v}}{dt}$ in time τ, $\boxed{v = \frac{F\tau}{m}}$
drift velocity term

$v = -\left(\frac{\tau}{m}\right)\nabla U$
↑ mobility
potential E gradient
↳ $\nabla U = \frac{dU}{dx}$

Difference Between Drift and Diffusion

Diffusion - Scattering from random motion
Drift - Scattering from external force

Bose Einstein Condensation

Macroscopic Quantum States

RE distr. $N(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)\beta T}-1} = V(\text{const.})(k_BT)^{3/2}\int_0^\infty d\varepsilon\cdot\varepsilon^{1/2}\frac{1}{e^{\beta\varepsilon+\alpha}-1}$

total number
$N = V\int D(\varepsilon)N(\varepsilon)d\varepsilon$
$D(\varepsilon)$ = $\varepsilon^{1/2}$ for particles w/ mass in 3D

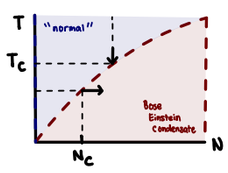
+ No ← condensate

$\beta = \frac{1}{k_BT}$
$\alpha = \frac{-\mu}{k_BT}$

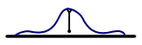
*classical limit: $\simeq e^{-(\beta\varepsilon+\alpha)}$

recall, for bosons:
$\mu \leq 0$ ← ground state energy
→ N has max value when $\alpha = 0$ $(\mu = 0)$
$N = N_{excited}(T,\mu) + N_0$
right at critical density $(N_0 = 0)$
$N = N_{excited}(T, \mu=0)$ "critical density"
$\sim T^{3/2}$

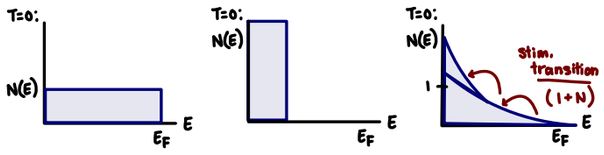
fix N, solve for T
$T_c = (\text{Const.})\left(\frac{N}{V}\right)^{2/3}$



- macroscopic # of particles
  ↳ in a single quantum state



⇒ number conservation, particles w/ mass, bosons @ high density



macroscopic wave

$K = \frac{2\pi}{\lambda}$
'quantized circulation'
'Persistant current'

$\psi(\varphi + 2\pi r) = \psi(\varphi)$
can satisfy with $N\lambda = 2\pi r$

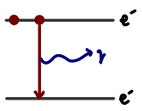
- as chemical potential $(\mu \to 0)$: excited state becomes more and more 'BEC'-like. [except @ ground state]
- Adding more particles can lower μ.
- only other way to force superconducting state: lower T

Liquid Helium
^4He - 2p, 2n, 2e
↳ integer spin: behaves like boson
Cooper pair - 2e

super current: carries electric current

$\Gamma_{i\to f} \propto (1+N_f)$ stimulation
$\Gamma_{i\to f} \propto (1-N_f)$ decoherence

Lasers - Light Amplification by stimulated emission radiation

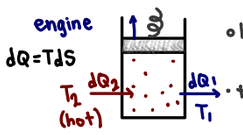


thermal cycles ("machines")

engine heat → work

refrigerator work → flow from cold to hot

engine
$dQ = TdS$



- heat flow from T_2 into system
  → raises pressure (S↑)
- turn off flow from T_2, piston rises
  → adiabatic, does work $(dW = PdV)$
- heat flow from system out T_1
  → allow piston to push back (adiabatic)

return to start

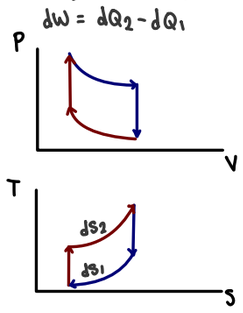
refrigerator
① Compress - T goes up
② Connect to cool bath T_2
→ lose heat
③ allow it to expand by doing work on piston
→ T drops below T_1
④ connect to T_1 → heat flow in

engine efficiency

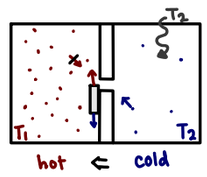
$\eta = \frac{dW}{dQ_2}$ ← work out / ← heat in $= \frac{dQ_2 - dQ_1}{dQ_2} = \frac{T_2dS_2 - T_1dS_1}{T_2dS_2}$

* if $dS_1 = dS_2$
↳ $\frac{T_2 - T_1}{T_2}$ 'carnot engine'

1st Law: $dQ_2 = dW + dQ_1$
$dW = dQ_2 - dQ_1$



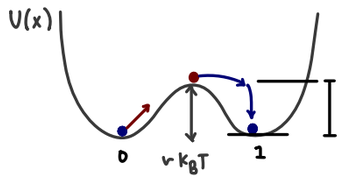
Maxwell's Demon - assume door costs negligible energy



$\Delta S < 0$

partial sol^n to paradox:
'demon' has entropy cost
↳ can be T_1
* information cost

model of "memory" - classical double well



- changing bit has minimum cost of k_BT (ΔU)

Information vs. entropy

recall, entropy: $S = k_B\log\Omega$ ≤ equiv. states

define,
shannon information: $I = \underbrace{\log_2\Omega_{i_3}}_{\eta \text{ bits}} - \log_2\Omega_{i_f}$

states that final state was determined from
states after measurement

- "actionable distinction" - ability to alter state based on available information causes actual state change

Szilard Engine - allow wall to be movable

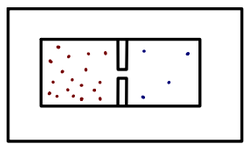
* information → work

looks like 100% efficiency

$\eta = \frac{k_BT_2}{k_BT_1}$ ← work done / ← heat

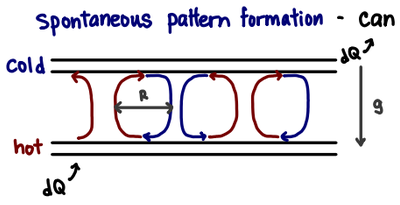
accounting for energy cost of demon at T_1
$\eta = \frac{k_BT_2 - k_BT_1}{k_BT_2}$
$= \frac{T_2 - T_1}{T_2}$

Operation of a machine follows 2nd Law?



machine itself exists in low entropy state

Spontaneous pattern formation - can machines appear spontaneously?

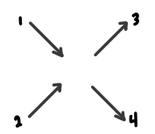


"convection cells"

'2nd Law doesn't apply in a closed system?'

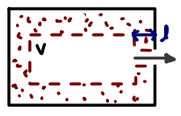
2nd Law - globally, $\Delta S > 0$

- locally, $\Delta S > 0$, w/ exception @ an interface



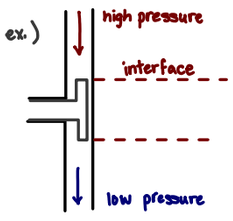
recall: H-Th^m. $\frac{dS}{dt} = -k_B\sum_{(\text{all states})}(N_1N_2 - N_3N_4)$

'detailed entropy increase' - each term in sum > 0



fraction that contributes $\propto \frac{Al}{V}$

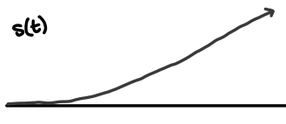
(transistor ex.)



Origin of life - animal cell, 30,000 machines

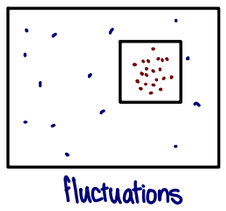
→ extremely low energy

Beginning of Universe: 2nd Law: $\Delta S > 0$



Big Bang $S \sim 10^{-100^{100}}$

Consciousness problem



Can calculate probability of spontaneous low entropy state

p(state) increases over larger regions

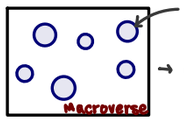
fluctuations

Philosophical Approaches to Entropy Problem

① positivism

② theism - first cause

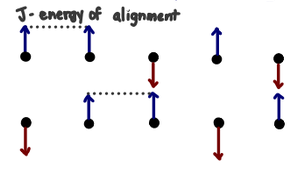
③ many-worlds of universes



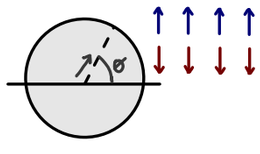
many subuniverses span all possibilities
→ anthropic principle
- observer / selection
• bubble of non-random (occam's razor)

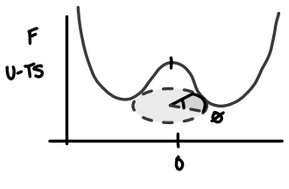
Spontaneous Symmetry Breaking

J - energy of alignment



$k_BT \gg J$ - random
$k_BT < J$ - aligned





no net spin - random